

Model Explainer: Equipment Failure Risk

Fleet-Wide Missed-Maintenance Prediction — Nairobi Depot Network

The Analogy

Think of this model like a car's check-engine light — but a smarter one that's watched thousands of engines and learned what "about to break down" actually looks like before the dashboard would normally warn you. It doesn't know anything about which specific car it's looking at, its make, or its history. It only watches how the engine is currently running: is output lower than it should be, and has that been true for a while, not just a one-off dip? When both are true together, it raises a flag. It's not a mechanic — it can't tell you exactly what's broken — but it's very good at telling you where to send the mechanic first.

The Drivers

Two features drive almost all of the model's decisions — see the chart below.

Feature	Importance
Rolling mean throughput (7-shift)	53.3%
Raw shift throughput	39.1%
Ambient temperature	4.2%
Everything else (voltage, throughput volatility, shift)	< 3.5% combined

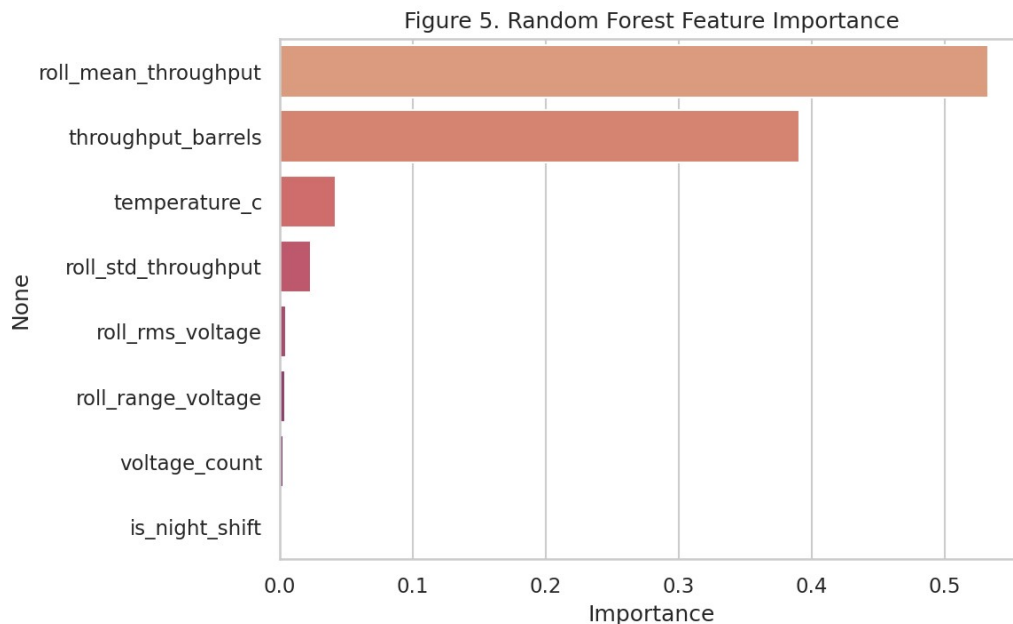


Figure 5. Random Forest Feature Importance (from week9_operational_ml.ipynb)

In plain English, the top 3 drivers are:

- 1. Sustained low output (53%) — the single biggest signal isn't one bad shift, it's a machine that's been running below its normal level for the better part of a week. A one-off dip doesn't trigger this; a pattern does.
- 2. Today's output (39%) — how the machine is performing right now, on top of the trend. Together with the rolling average, these two features alone account for over 90% of every decision the model makes.
- 3. Ambient temperature (4%) — a distant third. Weather-related conditions play only a minor supporting role; this model is fundamentally reading output, not the environment.

Trust & Limits

This model is a triage tool, not a verdict. It should guide which equipment a maintenance team looks at first — never replace a technician's judgment about what's actually wrong or whether to act.

Where it can go wrong:

- False positives (flagged, but actually fine): a temporary, explainable dip — a slow shift, a short pause for an unrelated reason — can look like sustained degradation for a few days before it self-corrects. Cost: an unnecessary inspection. Manageable, but repeated false alarms risk "alert fatigue," where the team starts tuning out real warnings too.
- False negatives (missed, but actually failing): this model was trained on one specific, clean failure pattern (a pump with a sharp, sustained output drop). A different kind of failure — intermittent, sudden, or one that doesn't show up in output first — could go undetected. The model has only ever seen one type of problem; it has no way to recognize a type it's never seen.

How the operations team should use this:

- Treat a "High Risk" flag as a prompt to inspect, not an automatic shutdown order.
- Pair it with what the crew already knows about the equipment — the model has no context on recent repairs, known quirks, or anything that isn't in the data.
- Track false positives and false negatives over time; if either climbs, that's a signal the model needs retraining on more recent or more varied failure data, not that the tool should be ignored.

Bottom line: this tool is very good at doing one specific, well-defined job — catching the exact failure pattern we've seen before, fast. It is not a general-purpose mechanic, and it shouldn't be trusted as one.